ASSIGNMENT 1
PROBABILITY & STATISTICAL DATA ANALYSIS
(SECI1143-03)

NAME	MATRIC NUMBER
MUHAMMAD AFIQ DANIAL BIN ROZAIDIE	A23CS0117
MOHAMED ALIF FATHI BIN ABDUL LATIF	A23CS0112
MUHAMMAD NAIM BIN ABDULLAH	A23CS0134
AHMAD ADIB ZIKRI BIN A.MAZLAM	A23CS0205

LECTURER: DR NOR HAIZAN BINTI MOHAMED RADZI

Question 1

a) Population is the entire collection of individual or object about which information is desired. In this case, all the university's students are the population.

Sample is a subset of the population, selected for study ~~is~~ in some prescribed manner. In this case, the university's students within Faculty of Computing are the sample.

b) i) Students' age
- discrete
- numerical
- nominal
- example: 19, 20, 21, 22

ii) Student's ethnicity
- discrete
- categorical
- nominal
- example: Malay, Chinese, Indian

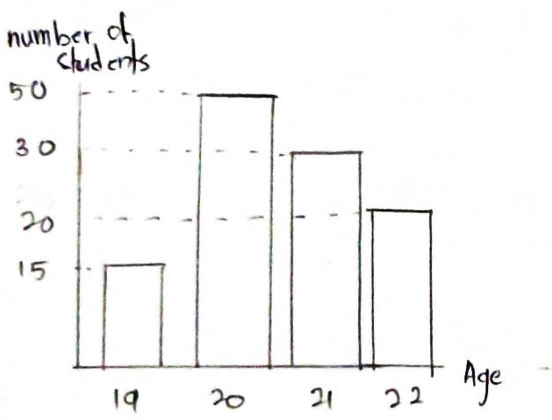
iii) Students' family income
- continuous
- numerical
- ratio
- example: RM 1000, RM 2000, RM 5000, RM 10000

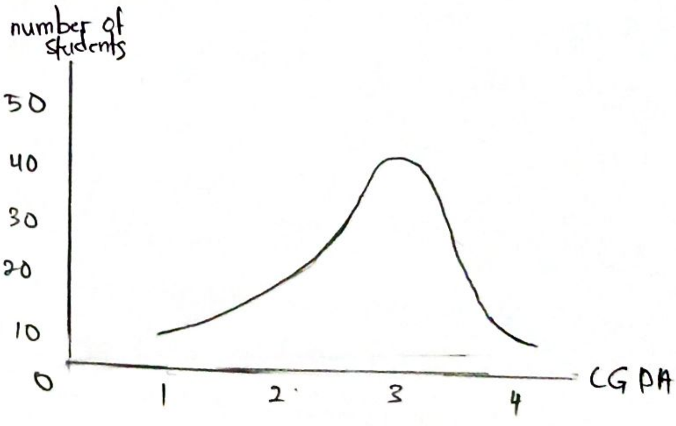
iv) Students' academic performance (CGPA)
- continuos
- numerical
- ratio
- example: 2.00, 3.00, 3.63, 4.00

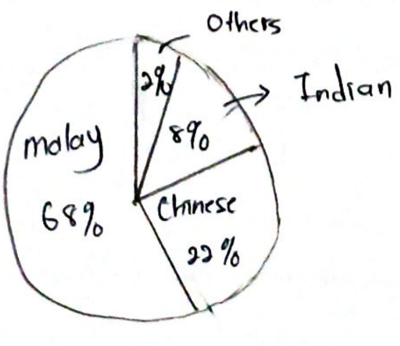
c) Primary data source
- Interview some students of Faculty of Computing. For example, ask them about their current CGPA.

d) Secondary data source
- Do research about students' current situation. For example, make a research about the average total income for ~~the~~ a single student's family.

e)







Question 2

number of student	frequency	relative frequency
0	3	$3/30 = 0.1$
1	2	$2/30 = 0.07$
2	7	$7/20 = 0.23$
3	7	$\frac{7}{20} = 0.23$
4	5	$5/20 = 0.17$
5	4	$4/20 = 0.13$
6	2	$2/30 = 0.07$
total	30	1

A packaged product has a label weight of 450 grams. The company goal is to fill each packages with at least 450 grams, but at most 458 grams. To check these goals , a random sample of 100 packages was selected and weighted, yielding the following weights, rounded to the nearest gram:

450	**450**	458	460	**450**	472	455	462	463	467
453	455	466	453	471	**451**	472	464	463	453
460	457	460	458	472	454	473	466	455	470
470	472	**452**	459	453	466	457	464	462	459
472	**450**	471	464	453	472	457	461	**450**	470
471	467	456	465	**450**	**450**	457	465	466	454
472	467	455	456	455	470	471	471	**450**	458
469	467	457	455	463	453	458	**452**	456	469
457	472	460	459	471	461	470	453	469	471
450	454	**451**	**450**	456	472	465	464	471	467

Construct :

a) Construct a histogram,

b) frequency polygon

c) ogive

(a)

w	Tabulation	f	w	Tabulation	f
450	卌 卌	10	465	III	3
451	II	2	466	IIII	4
452	II	2	467	卌	5
453	卌 II	7	468		0
454	III	3	469	IIII	4
455	卌 I	6	470	卌 I	6
456	IIII	4	471	卌 II	7
457	卌 I	6	472	卌 IIII	9
458	IIII	4	473	I	1
459	III	3			
460	III	3			
461	II	2			
462	II	2			
463	III	3			
464	IIII	4			

$$R = X_n - X_1$$
$$= 473 - 450$$
$$= 23$$

$$h = \frac{R}{i}$$
$$= \frac{23}{3.0089}$$
$$= 7.644$$
$$= 8$$

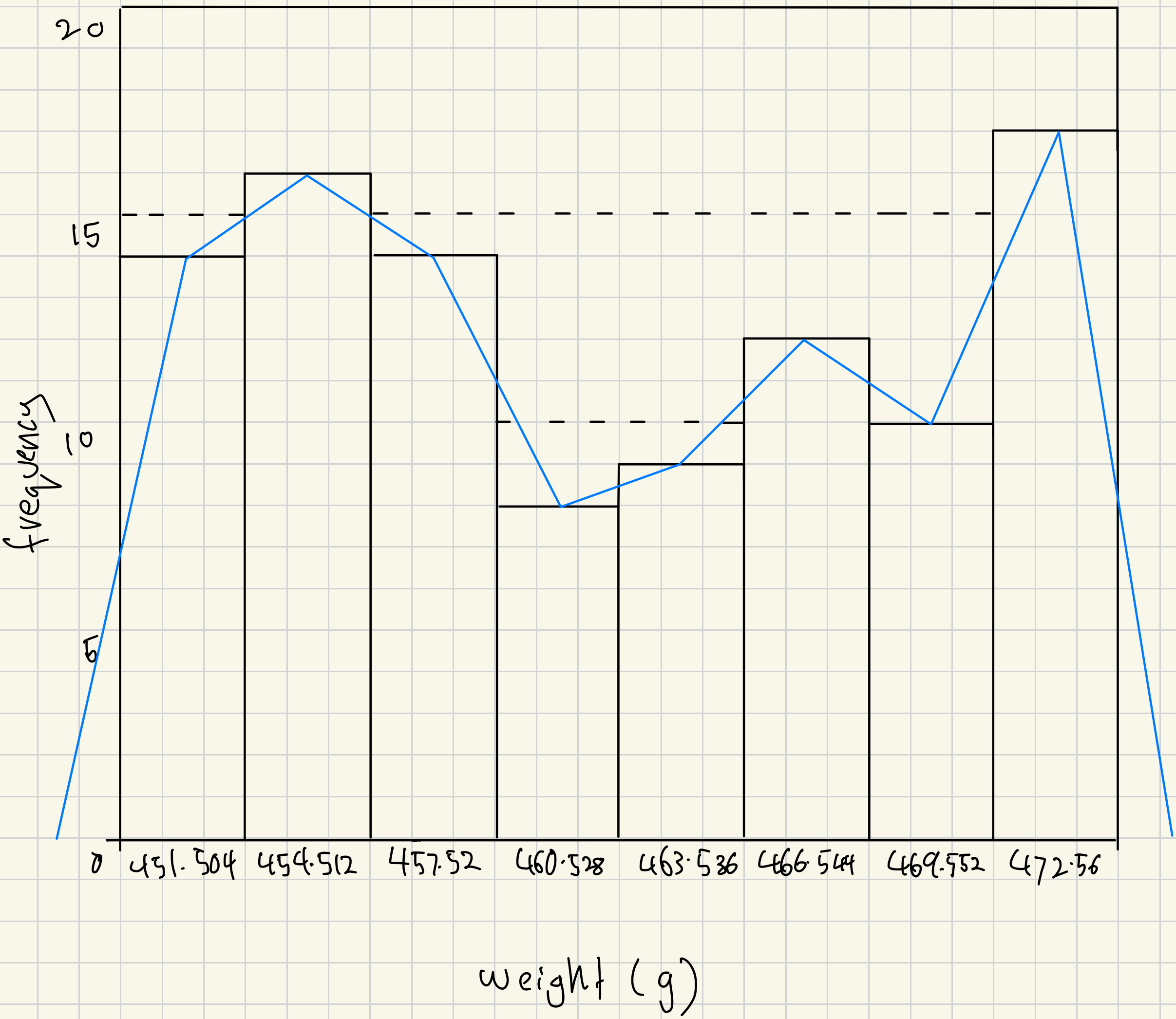
$$i = \frac{R}{1 + 3.322 \log n}$$
$$= \frac{23}{1 + 3.322 \log 100} = 3.0089$$

midpoint, $MP_1 = x_1 + \frac{i}{2}$
$$= 450 + \frac{3.008}{2}$$
$$= 451.504$$

midpoint
451.504
454.512
457.520
460.528
463.536
466.544
469.552
472.56

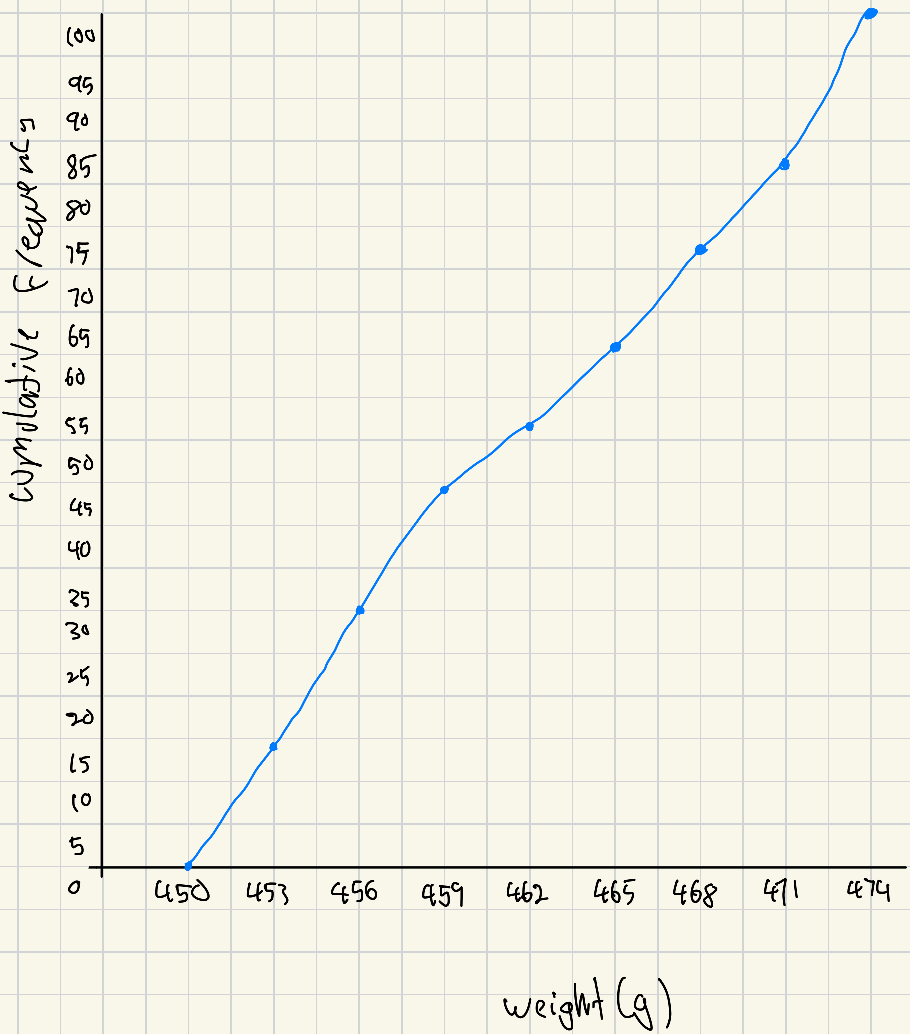
Cell Boundaries	Cell Midpoint	f
450 - 453	451.504	14
453 - 456	454.512	16
456 - 459	457.520	14
459 - 462	460.528	8
462 - 465	463.536	9
465 - 468	466.544	12
468 - 471	469.552	10
471 - 474	472.56	17

Histogram & Polygon



Ogive

Cell Boundaries	Cell Midpoint	f	cf
450 - 453	451.504	14	14
453 - 456	454.512	16	30
456 - 459	457.520	14	44
459 - 462	460.528	8	52
462 - 465	463.536	9	61
465 - 468	466.544	12	73
468 - 471	469.552	10	83
471 - 474	472.56	17	100



Question 4

78, 702, 765, 811, 832, 855, 896, 902, 905, 918, 919, 920, 923, 929, 936, 938, 948, 950, 956, 958, 958, 970, 972, 978, 1009, 1009, 1022, 1035, 1037, 1045, 1067, 1085, 1092, 1102, 1122, 1126, 1151, 1156, 1157, 1157, 1162, 1170, 1195, 1195, 1196, 1217, 1237, 1311, 1333, 1340.

$N = 50$

$P = 25, i = \frac{25}{100} \times 50$

$= 12.50$

$k = 13$

$Q_1 = 923$

$P = 50, i = \frac{50}{100} \times 50$

$= 25$

$k = 25$

$Q_2 = \frac{Y[25] + Y[26]}{2}$

$= \frac{1009 + 1009}{2}$

$= 1009$

$P = 75, i = \frac{75}{100} \times 50$

$= 37.50$

$k = 38$

$Q_3 = 1156$

$IQR = Q_3 - Q_1$

$= 1156 - 923$

$= 233$

Upper limit $= Q_3 + (1.5 \times IQR)$

$= 1156 + (1.5 \times 233)$

$= 1505.50$

Lower limit $= Q_1 - (1.5 \times IQR)$

$= 923 - (1.5 \times 233)$

$= 573.50$

Max = 1340

Min = 702

Outliners = 78